



Estimating Domestic Passenger Demand for Airlines in South Korea

Based on Random Coefficients Logit Model

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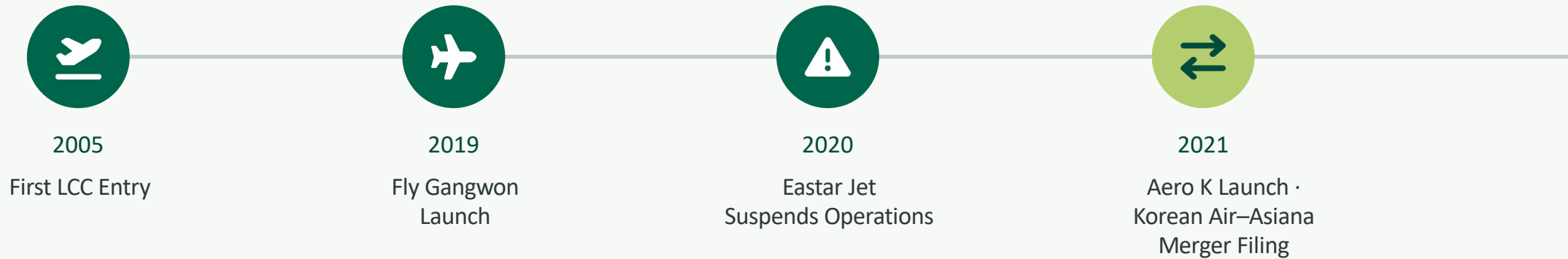


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Conclusion



Research Purpose



2
Domestic Airlines
before 2005



8
Domestic Airlines
as of Dec. 2020

RESEARCH OBJECTIVE

Estimate domestic passenger demand functions
to understand airline choice and demand characteristics



Research Questions

01

Route-Level Own-Price Elasticity

Estimate demand functions separately for Jeju and inland routes using a random coefficients logit model, and test whether inland demand is more price elastic.

02

Impact of LCC Growth

Compare the estimates with prior studies before LCC entry to examine how LCC entry and growth changed price elasticity in domestic passenger demand.

03

Substitution across Airlines

Examine cross-price elasticities to explore whether substitution among FSCs differs from substitution between FSCs and LCCs.



Sample & Scope

All domestic routes | Excluding stopover and international-transfer routes | 9 domestic airlines



14 Airports Nationwide

7 International Airports

- Gimpo
- Yangyang
- Cheongju
- Muan
- Daegu
- Gimhae
- Jeju

7 Domestic Airports

- Wonju
- Gunsan
- Yeosu
- Gwangju
- Pohang
- Sacheon
- Ulsan



9 Domestic Airlines

2 FSCs

- Korean Air
- Asiana Airlines

7 LCCs

- Jin Air
- Air Busan
- Air Seoul
- Jeju Air
- T'way Air
- Fly Gangwon
- (Estar Jet)



Data Period & Market Definition

Period

January 2016 – December 2020 (monthly)

Market Definition

Define routes by origin–destination pairs,
and set each route × month as one market.



14

Airports



9

Airlines



2,086

Markets



5,757

Observations

Geographic scope: all domestic routes · excluding stopover and international-transfer routes · 9 domestic airlines



Methodology: Discrete Choice Model



Why a Discrete Choice Model?

- Suitable for estimating demand for differentiated products.
- Allows individual heterogeneity.
→ Models differ depending on which parameters capture heterogeneity.

Simple Logit

Random Coefficients Logit

Utility when consumer i chooses product j in market t

$$u_{ijt} = \alpha_i p_{jt} + X_{jt} \beta_i + \xi_{jt} + \epsilon_{ijt}$$

$$i = 1, \dots, I_t, \quad j = 0, \dots, J, \quad t = 1, \dots, T$$



p_{jt}

Price Index



X_{jt}

Observed Product Characteristics



ξ_{jt}

Unobserved Product Characteristics



ϵ_{ijt}

i.i.d. Type I extreme value error term



Simple Logit vs. Random Coefficients Logit

1 Simple Logit Model

- $\alpha_i = \alpha, \beta_i = \beta$
- Only ϵ_{ijt} captures consumer heterogeneity



Limitation: IIA
(Independence of Irrelevant Alternatives)

2 Random Coefficients Logit

- $\alpha_i, \beta_i, \epsilon_{ijt}$ all capture individual heterogeneity
- Relaxes IIA
- Can directly incorporate demographic information such as income.

$$\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma v_i, \quad v_i \sim P_v, \quad D_i \sim P_D$$



Key Variables I



Price Index

Weighted average of weekday, weekend, flexible, and peak-season fares using flight frequency weights

Korea Airports Corporation · MOLIT report · news articles



Passengers

Paid passengers by airline and route

Korea Airports Corporation aviation statistics



Flight Frequency

Monthly flight frequency

Korea Airports Corporation aviation statistics



Outside good

Route-level users estimated from KTX/SRT station boarding and alighting counts

KTX/SRT passenger transport statistics



Route Distance

Straight-line distance between origin and destination airports (m)

Kakao Open API



High-Speed Rail Routes

Number of direct KTX/SRT routes between nearest stations within 30 km of each airport

LetsKorail · SRT websites



Key Variables II



Airport Dummies

14 airport dummies
(Reference: Gwangju)



Airline-Type Dummies

FSC (reference) · Air Seoul · Fly Gangwon ·
Other LCCs



Overall LCC Dummy

FSC = 0 · LCC = 1



Instrumental Variable

Number of FSCs among other airlines
operating in the market



Log Income per Capita

Previous-year average monthly household income
÷ household members (unit: KRW 10,000)

Household Finance and Welfare Survey
& KLIPS waves 19–22



Unobserved Individual Characteristics

Random draws from the standard
normal distribution



Price Index Construction

Weighted average of weekday, weekend, flexible, and peak-season fares



Inland Routes	$r_{peak}P_{peak} + (1 - r_{peak})(r_{weekday}P_{weekday} + r_{weekend}P_{weekend})$
Arriving Jeju	$r_{peak}P_{peak} + (1 - r_{peak})(r_{weekday}P_{weekday} + r_{frisat}P_{flexible} + r_{sun}P_{weekend})$
Departing Jeju	$r_{peak}P_{peak} + (1 - r_{peak})(r_{weekday}P_{weekday} + r_{frisat}P_{weekend} + r_{sun}P_{flexible})$

Condition : $r_{weekday} + r_{weekend} = r_{weekday} + r_{frisat} + r_{sun} = 1$



Estimation Results: Linear Parameters (Inland)

Variable	Model 2-1 Inland	Model 2-2 Jeju
Price Index	-0.00014* (0.00007)	-0.000061 (0.000077)
Flight Frequency	0.00614*** (0.00024)	0.004738*** (0.000088)
High-Speed Rail Routes	-0.22449*** (0.07390)	
Route Distance	0.00002*** (0.00000)	
Air Seoul	0.6689*** (0.071)	
Fly Gangwon	0.5985*** (0.021)	
Other LCCs	-0.2893*** (0.074)	
Overall LCC		0.3101*** (0.000)

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Interpretation



Fare + KRW 1,000

Utility decreases by 0.14



Flights +10

Utility increases by 0.0614



High-Speed Rail Route +1

Air travel utility decreases by 0.224



Fly Gangwon

Only operates monopoly routes
→ positive coefficient



Other LCCs

Negative coefficient
→ possibly due to in-flight service differences



Estimation Results: Linear Parameters (Jeju)

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Robust standard errors in parentheses | *** p<0.01, ** p<0.05, * p<0.1

Interpretation



Fare + KRW 1,000

Utility decreases by 0.061



Flights +10

Utility increases by 0.047



Overall LCC : significant positive coefficient

Possibly driven by customer promotions, corporate benefits, and annual memberships



Estimation Results: Nonlinear Parameters

Variable	Inland Routes	Jeju Routes
Log Income	0.000004 (0.00038)	0.000005*** (0.000000)
Price × Log Income	0.00002* (0.00001)	0.000010 (0.000014)
Unobserved Individual Characteristics	0.46959*** (0.00707)	0.411874*** (0.000000)
Price × Unobserved Individual Characteristics	0.0000007 (0.00001)	0.000001 (0.000012)
Observations	1,358	4,399



Inland Routes

Higher-income consumers are relatively price inelastic



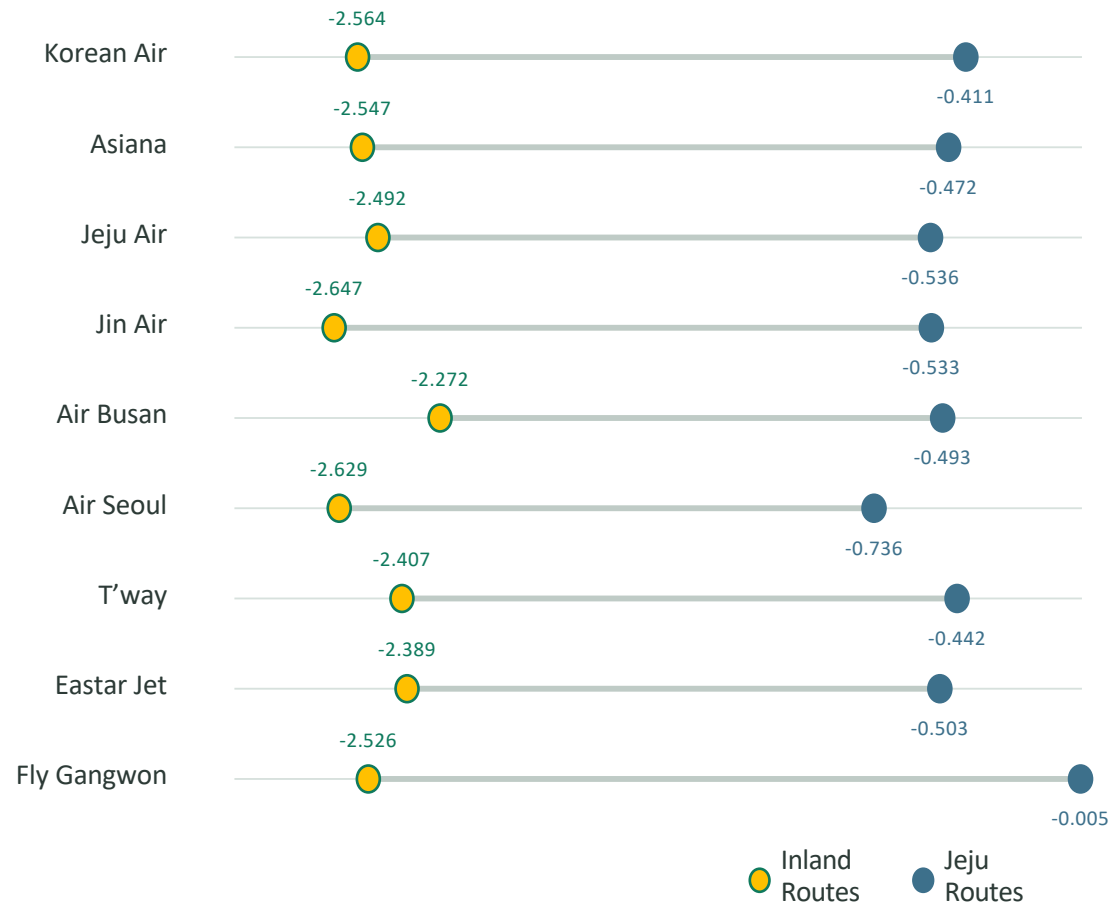
Jeju Routes

Higher-income consumers derive greater utility

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1



Own-Price Elasticity by Airline



Average Own-Price Elasticity

Inland Routes

-2.499

Previous Literature
vs -1.24

Jeju Routes

-0.474

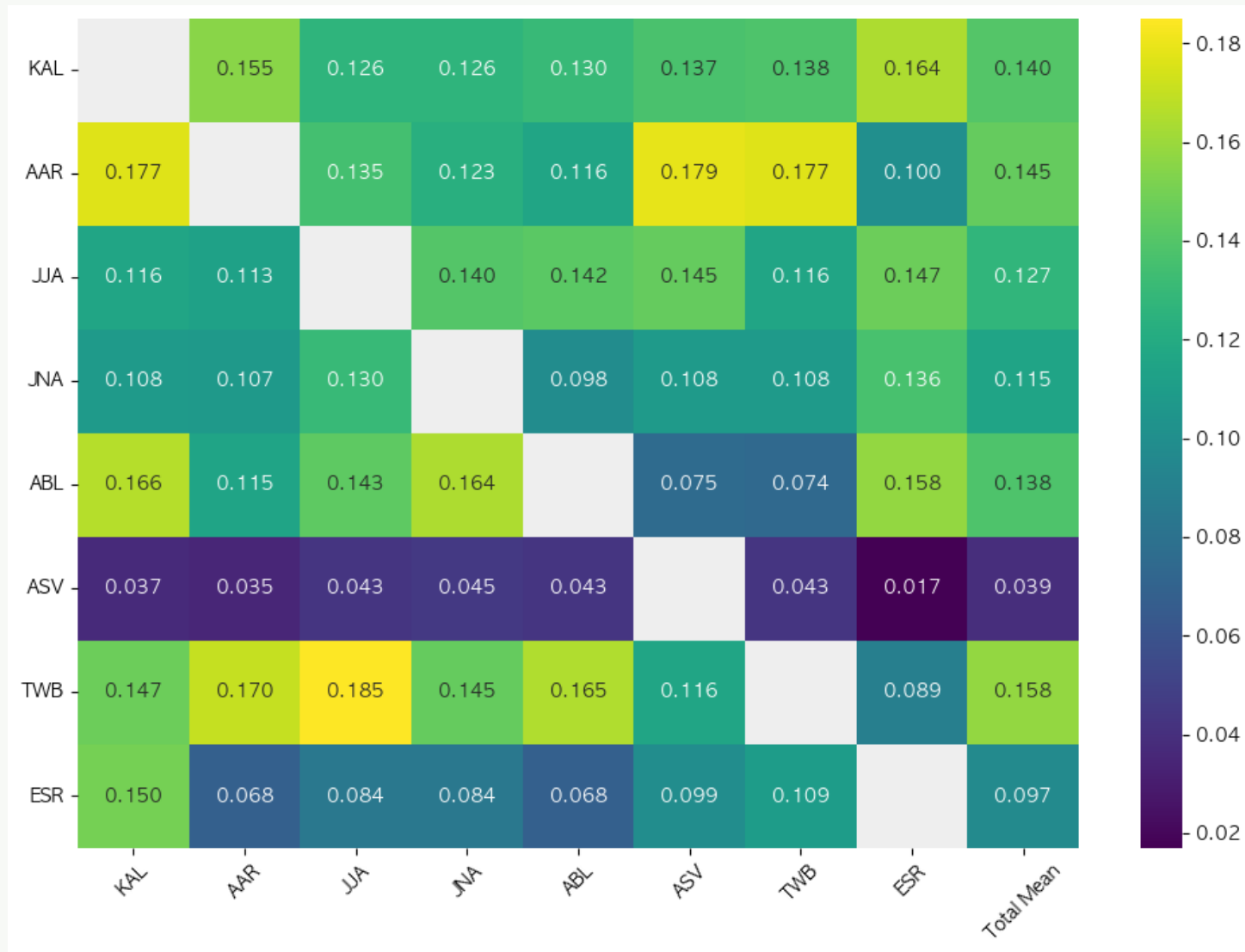
Previous Literature
vs -0.25

- Inland routes are more elastic.
- Elasticity increased relative to prior studies before LCC entry.

Cho & Jung (2006)



Cross-Price Elasticity: Jeju Routes





Cross-Price Elasticity: Inland Routes





Summary of Findings

01

Route-Level Price Elasticity

Average own-price elasticity is -2.499 for inland routes and -0.474 for Jeju routes.

This is consistent with the expectation that Jeju route demand is less price elastic than inland route demand.

02

Effect of Competition

Compared with prior studies, price elasticity increased for both Jeju and inland routes after LCC entry.

This can be interpreted as evidence of increased competition from LCC entry and growth.

03

Substitution among FSCs

On most routes, substitution between FSCs was higher than substitution between FSCs and LCCs, but the gap was not large. For inland routes, Asiana's cross-price elasticity with respect to Korean Air's fare changes was lower than that of other airlines.



Contribution & Limitations



Contribution

To the author's knowledge, this is the first attempt to estimate the domestic passenger demand for airlines in South Korea by applying the random coefficients logit model.



Extension

Incorporating a supply-side model would enable the analysis of equilibrium outcomes in the airline industry following structural changes such as mergers.



Limitation

Because the price index is based on published fares, it may not fully reflect actual transaction prices and consumer choices.



Future Improvement

Future research could account for additional unobserved characteristics.



Thank you.

Estimating Domestic Passenger Demand for Airlines · Jungmin Yoo